

Appendix A2 - South-west Corner Marine Park (Western arm) Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	Length
					Count	Count	
Jun 2020	2020-06 south-west stereo-BRUVs	BRUV	NPZ	42	40	39	33
Oct-Nov 2020	2020-10 south-west stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	275	273	272	244
Oct-Nov 2020	2020-10 south-west BOSS	BOSS	Coastal waters, NPZ, SPZ	262	262	-	-
Mar 2021	2021-03 West-Coast BOSS	BOSS	Coastal waters, NPZ	154	154	-	-
Mar 2023	2023-03 Augusta stereo-BOSS	BOSS	Coastal waters, NPZ, SPZ	73	70	-	-
Mar 2023	2023-03 SwC BOSS	BOSS	NPZ	88	88	-	-
Mar 2023	2023-03 SwC stereo-BRUVs	BRUV	NPZ	32	32	32	32
Oct 2024	2024-10 SWC stereo-BRUVs	BRUV	Coastal waters, NPZ, SPZ	190	190	188	178
Nov 2024	2024-11 SwC stereo-BOSS	BOSS	Coastal waters, MUZ, NPZ, SPZ	137	117	-	-
Mar-Apr 2025	2025-04 South-west Corner BOSS	BOSS	NPZ, SPZ	210	210	-	-
Apr 2025	2025-04 South-west-corner stereo-BRUVs	BRUV	NPZ, SPZ	150	148	143	137

Benthic ecosystem extent

Observations

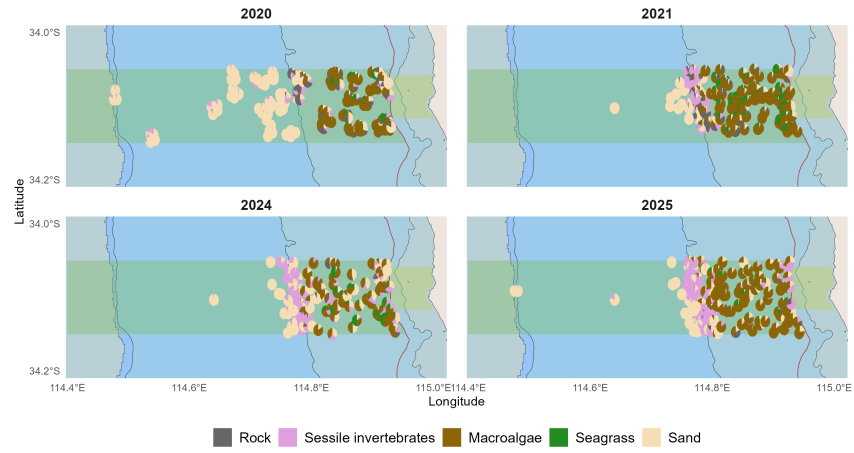


Figure A2 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

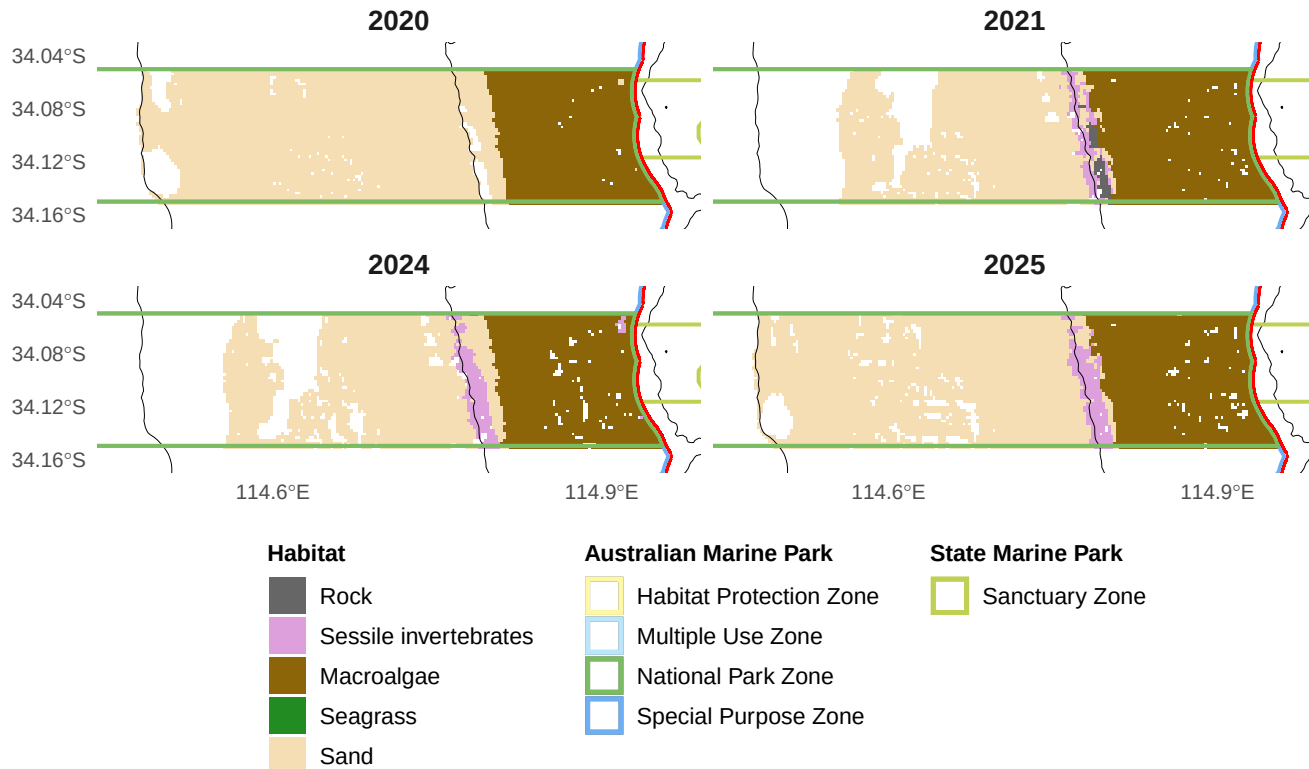


Figure A2 2: Predicted dominant habitat type by year from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

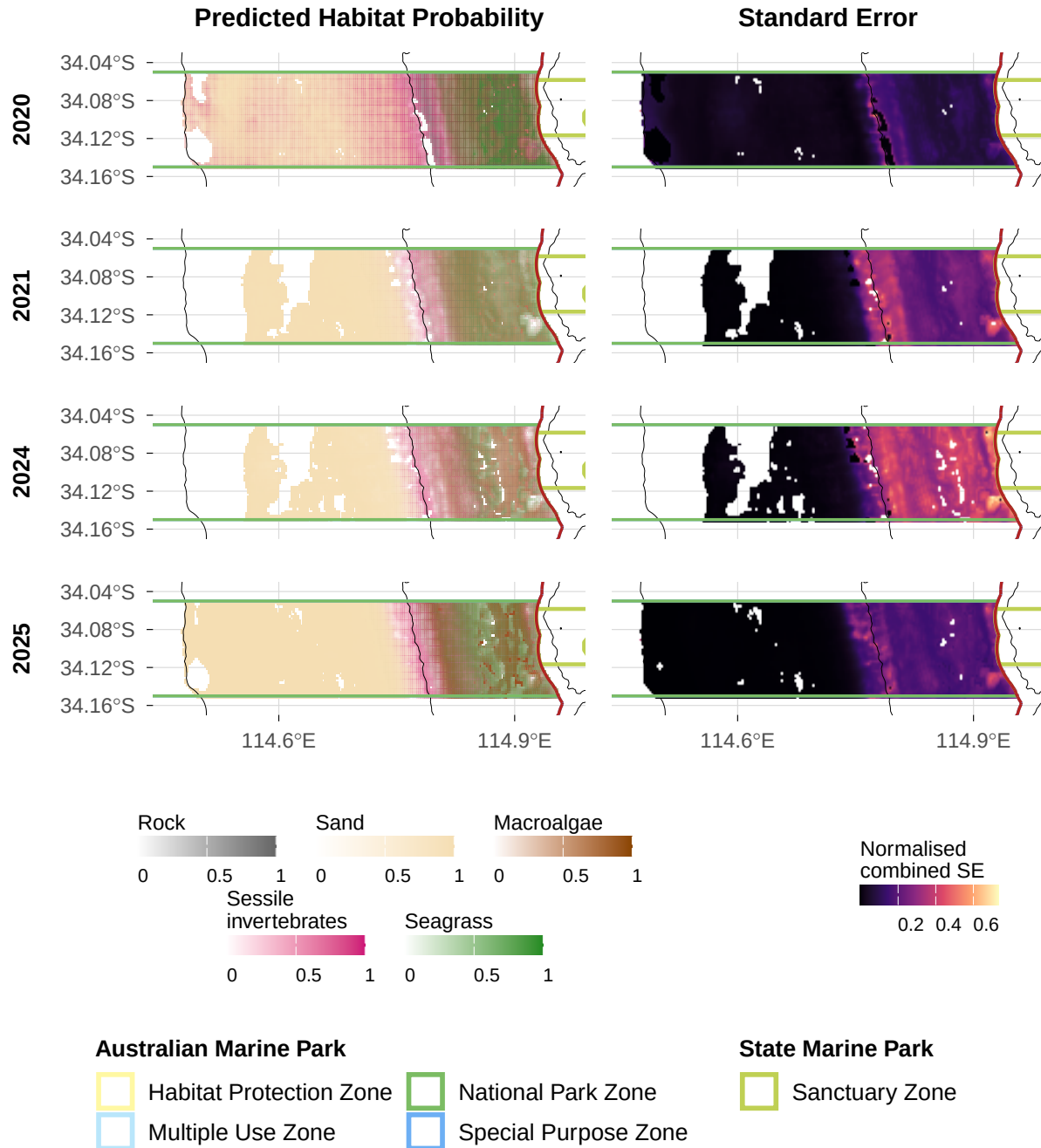


Figure A2 3: Predicted habitat distribution probability and uncertainty by year from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

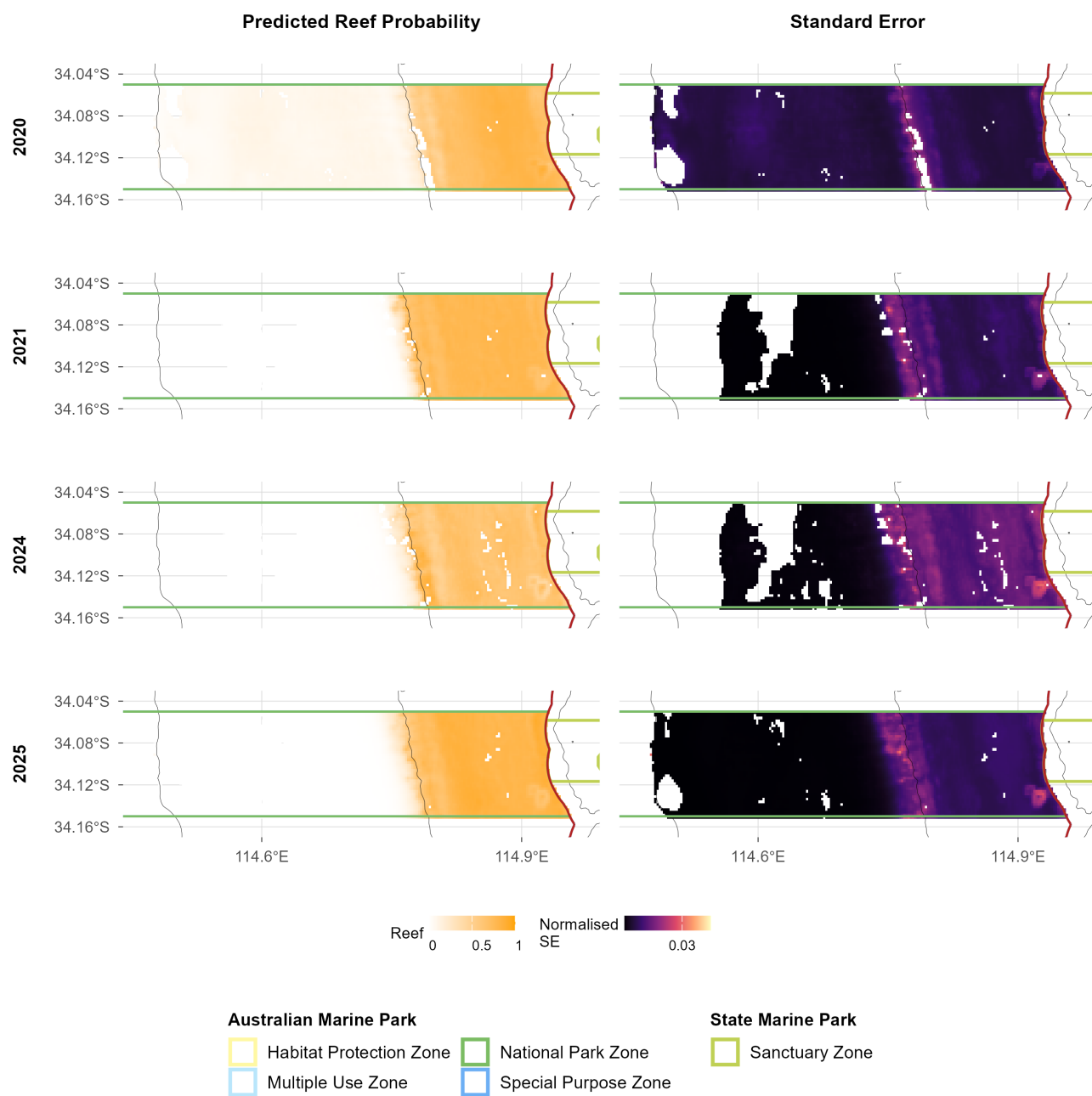


Figure A2 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

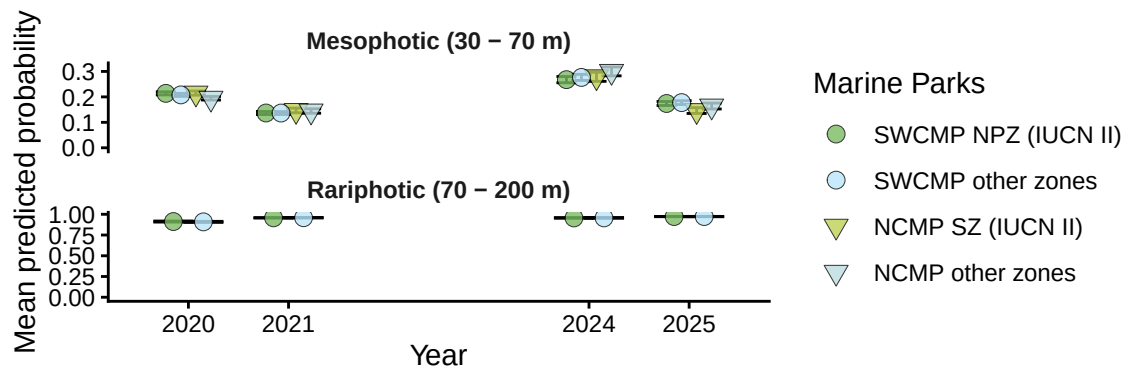


Figure A2 5.1: Sand: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWCMP = South-west Corner Marine Park (Commonwealth), NCMP = Ngari Capes Marine Park (State).

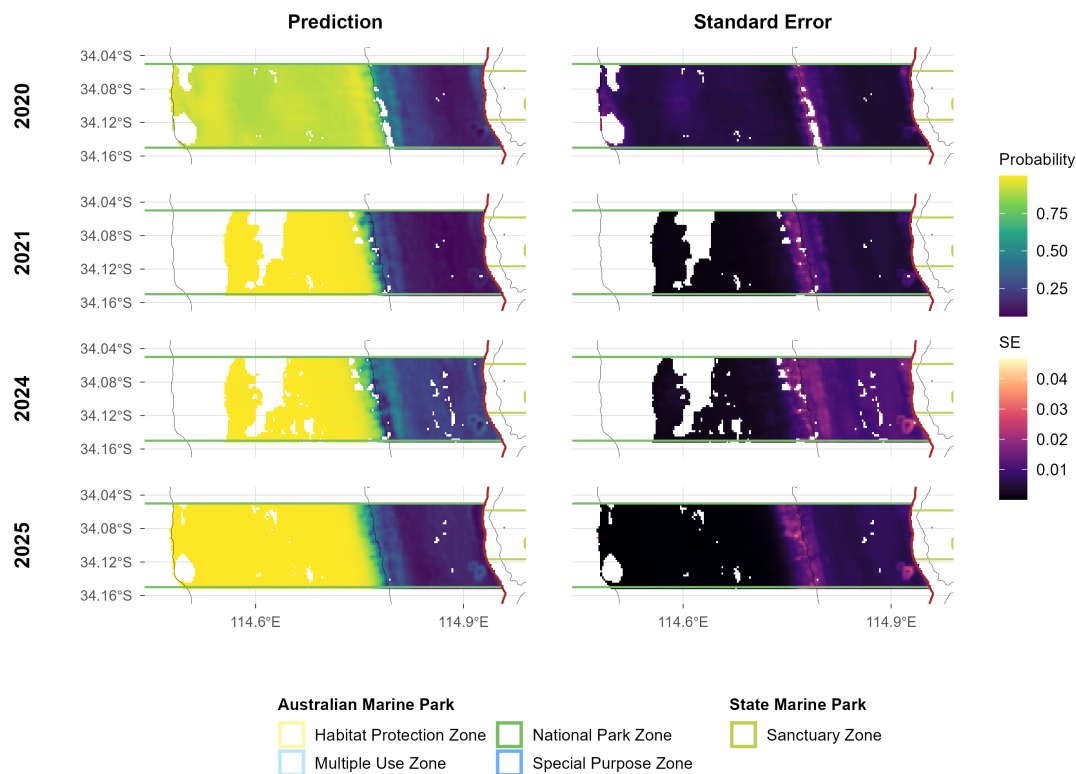


Figure A2 5.2: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

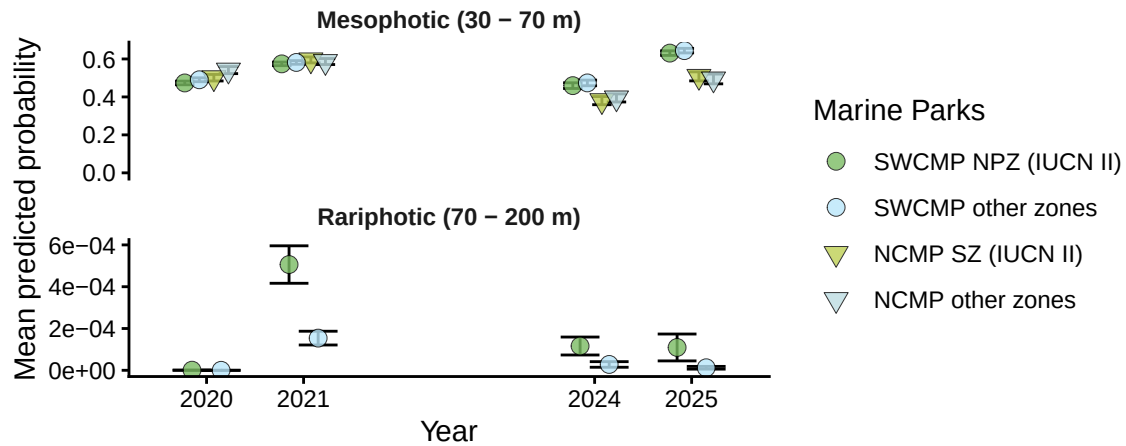


Figure A2 6.1: Macroalgae: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWCMP = South-west Corner Marine Park (Commonwealth), NCMP = Ngari Capes Marine Park (State).

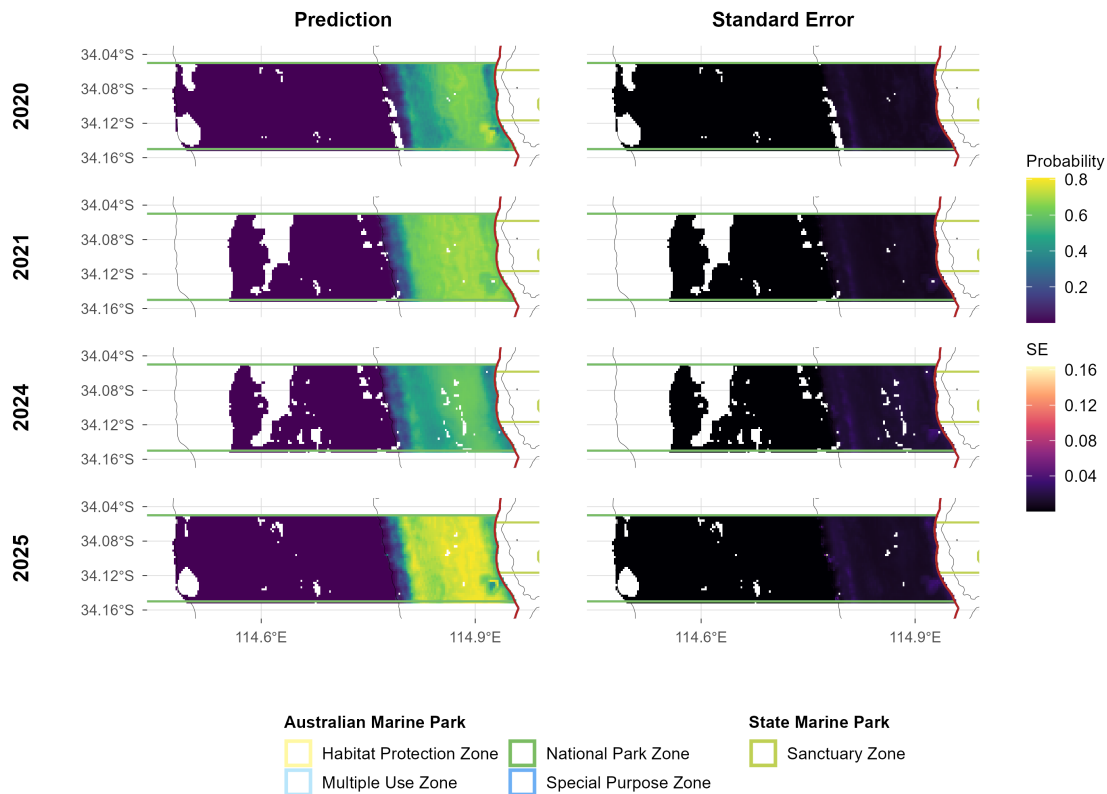


Figure A2 6.2: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Seagrass

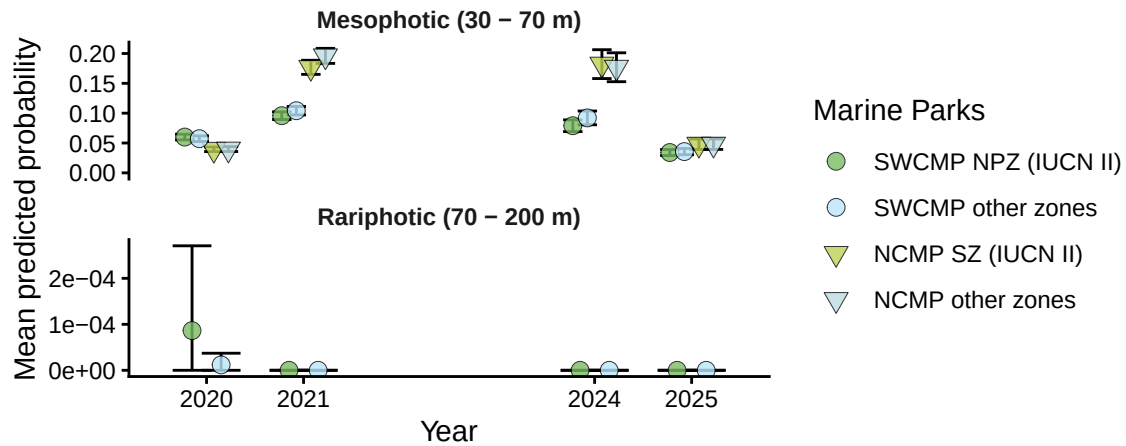


Figure A2 7.1: Seagrass: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWCMP = South-west Corner Marine Park (Commonwealth), NCMP = Ngari Capes Marine Park (State).

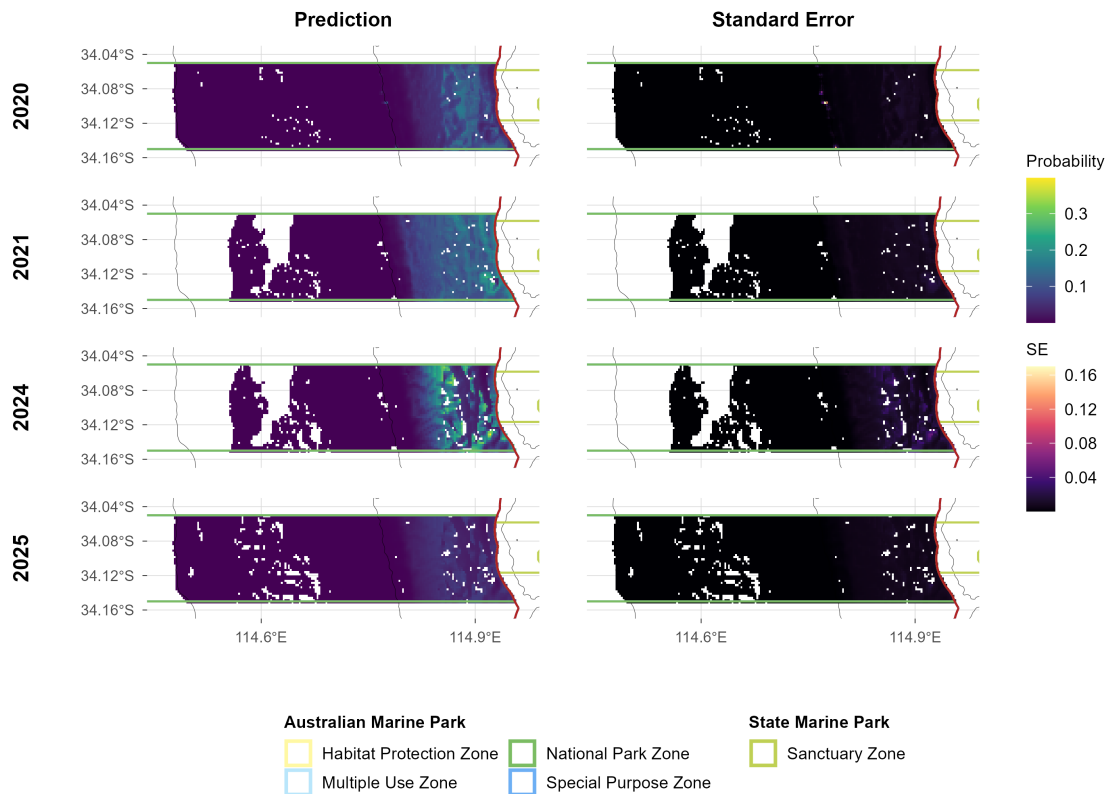


Figure A2 7.2: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

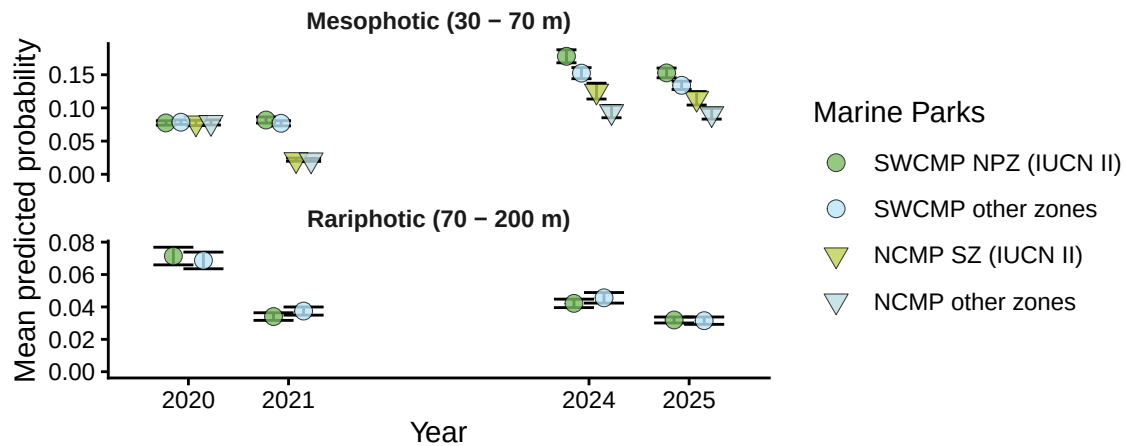


Figure A2 8.1: Sessile invertebrates: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWCMP = South-west Corner Marine Park (Commonwealth), NCMP = Ngari Capes Marine Park (State).

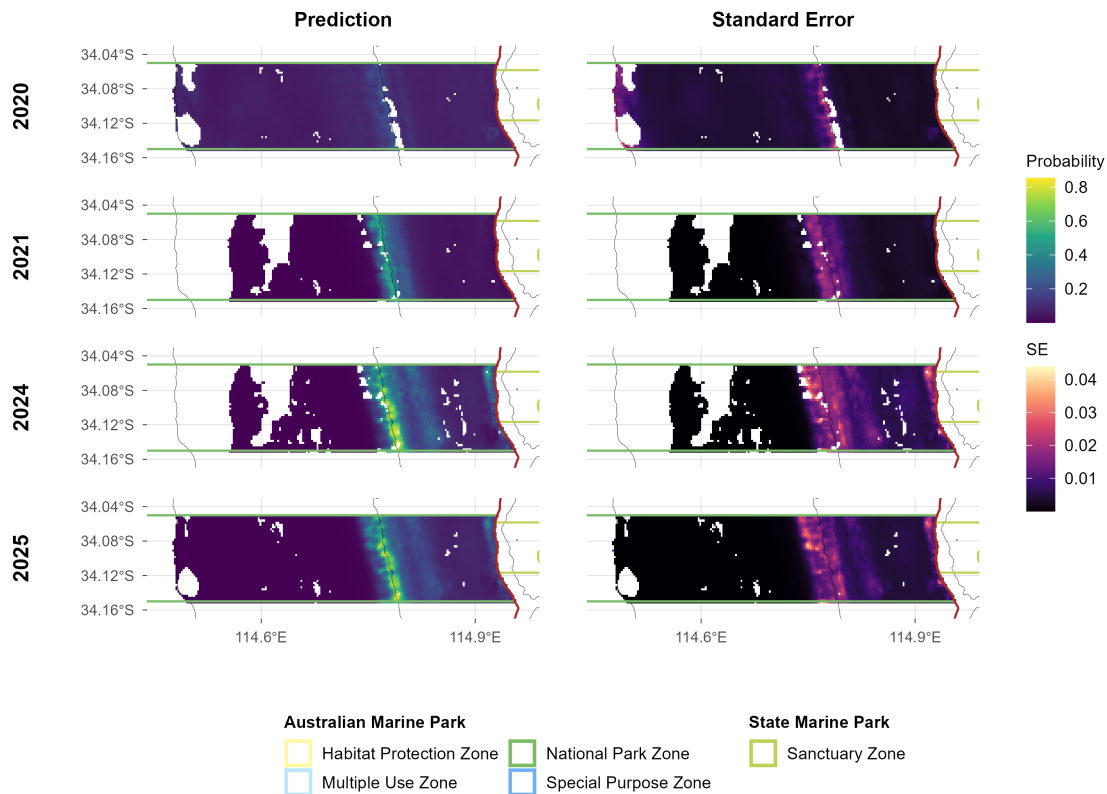


Figure A2 8.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Rock

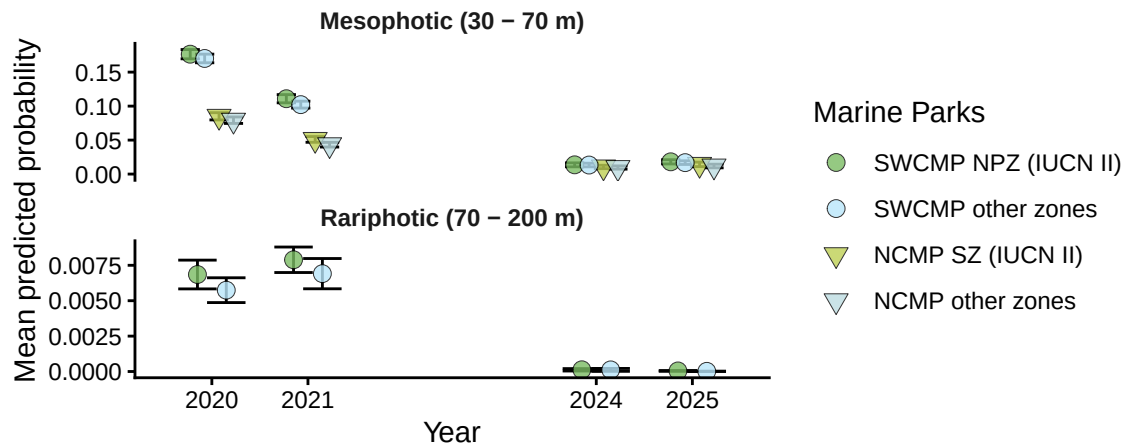


Figure A2 9.1: Rock: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWCMP = South-west Corner Marine Park (Commonwealth), NCMP = Ngari Capes Marine Park (State).

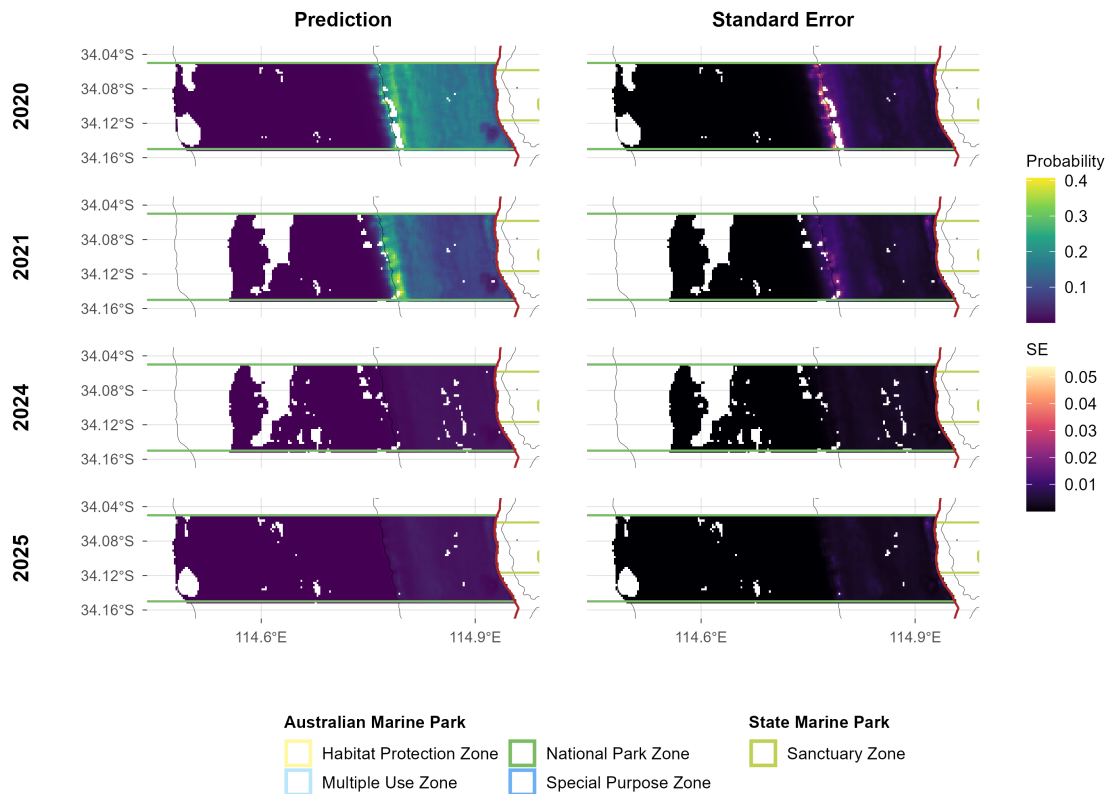


Figure A2 9.2: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.